

Target-Driven Subgroup Discovery for Static and Longitudinal Data with Interpretable Decision-Forest Explanations

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Abstract

Subgroup discovery seeks human-interpretable descriptions of subsets of data whose target values are exceptional relative to the overall population. We study a *target-driven* formulation in which “interesting” instances are identified using the target column alone and are subsequently characterized using compact decision-forest rules. We evaluate two target-only subgrouping strategies on static tabular datasets: (i) single-linkage clustering on the target with a high-quantile override and (ii) a SyFLoW-inspired bootstrap procedure that proposes target ranges and scores them using size-regularized KL divergence with diversity-aware selection. To address longitudinal datasets, we convert country–date panels into subgroup-ready tables using transparent temporal feature construction. Beyond basic lags and rolling statistics, we expand the decision-tree *question vocabulary* with multi-horizon lags, deltas, growth rates (CAGR), local regime-shift indicators, and threshold-crossing times. We additionally study a longitudinal target formulation where the subgrouping target is the local trend (slope) of life expectancy rather than its level. We select a small, interpretable forest by searching across randomized forest configurations and choosing a high-accuracy, low-question solution; all trees are exported for auditability. Finally, we report an XGBoost baseline with time-aware cross-validation to quantify the predictive signal present in temporal features.

1. Introduction

Subgroup discovery (SD) aims to identify subsets of a dataset whose target distribution differs meaningfully from

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that of the full population, typically under interpretability constraints (e.g., rule lists or bounded-depth decision trees). While many SD methods directly search feature space for descriptive rules, an alternative and practically useful viewpoint is *outcome-first subgrouping*: first identify exceptional behavior in the target, then explain it using interpretable patterns in the features. This viewpoint aligns with applied workflows where analysts first observe unusual outcomes (e.g., unusually high costs or atypical public-health trajectories) and then seek compact explanations.

This paper develops and evaluates a modular SD pipeline that supports both (i) static tabular data and (ii) longitudinal panel data. Our design separates the workflow into: (a) *target-driven subgroup labeling*, (b) *longitudinal feature construction* (for panel data), and (c) *interpretable rule extraction* via compact decision-forest explanations. The outcome is a reproducible, end-to-end approach that yields (1) labeled interesting instances and (2) a small set of human-readable decision rules describing them.

Contributions.

- We formalize a target-driven SD pipeline that produces subgroup labels using only the target column and then learns *interpretable* decision-forest explanations from features.
- We extend the pipeline to longitudinal data via standard, transparent temporal feature engineering, enabling SD in country–date panels.
- We provide a detailed experimental specification (including all hyperparameters) and report results on static and longitudinal datasets, alongside a time-aware boosted-tree baseline.

Scope and positioning. We do *not* claim novelty in temporal feature engineering; rather, we focus on an integrated SD workflow where (i) subgroup *identification* is target-driven and (ii) subgroup *characterization* is interpretable and feature-based. This distinction is important: our procedure explains target-defined “interesting” instances rather than performing direct feature-space subgroup search.

2. Problem Setup

Let a dataset be (X, y) with N instances, where $X \in \mathbb{R}^{N \times d}$ are features and $y \in \mathbb{R}^N$ is a real-valued target. We consider the following two-stage formulation.

Definition 2.1 (Target-driven subgroup labeling). A target-driven labeling method is a mapping $g : \mathbb{R}^N \rightarrow \{0, 1\}^N$ that assigns each instance a binary label $z_i = g(y)_i$, where $z_i = 1$ denotes an “interesting” instance.

Definition 2.2 (Interpretable subgroup explanation). Given labeled data (X, z) , an explanation model produces a small collection of rules (e.g., decision trees) whose predictions approximate z while remaining compact and interpretable.

Longitudinal setting. For panel data, each entity e (e.g., a country) has observations over time t . We denote raw observations as $(e, t, X_{e,t}, y_{e,t})$. We transform this panel into a tabular dataset $(\tilde{X}, y)$ via temporal feature engineering so that the same labeling and explanation components can be reused.

3. Method

3.1. Target-driven labeling on static data

We use two labeling strategies. Both operate on y only and output $z \in \{0, 1\}^N$.

(A) Original: clustering + high-quantile override. We apply single-linkage hierarchical clustering on the scalar target y . We identify the largest cluster (dominant mass) and label all other clusters as interesting. Additionally, we mark instances with y above a high quantile q as interesting to ensure extreme high values are always captured. This produces a conservative “mostly-normal vs. exceptional” split.

(B) SyFLoW-inspired: bootstrap target ranges with KL scoring. We repeatedly bootstrap the target values and cluster each bootstrap sample. Each cluster yields a candidate target range $[c_{\min}, c_{\max}]$ which maps back to a mask over the original data. We score each candidate by a size-regularized distributional divergence:

$$\text{quality}(m) = \left(\frac{|m|}{N}\right)^\beta \cdot \text{KL}(p(y | m) \parallel p(y)), \quad (1)$$

where KL is computed via histogram estimates using global binning. We then greedily select a diverse set of candidates by adding a diversity term based on pairwise KL and suppressing near-duplicates using a Jaccard-overlap constraint. The union of selected masks defines z .

3.2. Longitudinal data construction

To enable subgroup discovery in longitudinal datasets, we convert a panel into a “subgroup-ready” table using explicit temporal descriptors. For each entity e and time t with sufficient history, we compute a feature dictionary that expands the set of decision-tree questions available to the explanation module.

Temporal feature dictionary (question vocabulary). For the target and for each numeric indicator, we compute:

- **Multi-horizon lags:** lag-1, lag-3, lag-5.
- **Rolling statistics:** rolling mean and rolling standard deviation over a window of length w .
- **Local trend:** slope of a least-squares line fit over the last w observations.
- **Deltas:** $\Delta_1 = x_t - x_{t-1}$ and $\Delta_5 = x_t - x_{t-5}$ (when available).
- **Growth rates:** CAGR_5 and CAGR_{10} computed over the last 5 and 10 observations (when defined).
- **Regime shift:** an indicator for sign changes between recent and previous local trends (e.g., $\text{trend}_{\text{last5}} \cdot \text{trend}_{\text{prev5}} < 0$ when both are defined).

These features allow interpretable questions such as: “Is recent growth positive?”, “Did the trend reverse in the last decade?”, or “Is volatility high in the recent window?”

Threshold-crossing time features. Optionally, for a small set of target thresholds, we add the first year the target crosses above or below each threshold (e.g., “first year life expectancy exceeds 70”). In our WBD experiments we use thresholds $\{50, 70\}$.

Target mode (level vs. trend). In addition to using the target value $y_{e,t}$ itself, we also consider a *trend-target* formulation in which the subgrouping target is the local slope of the target series computed over the same rolling window. This enables outcome-first subgrouping over *dynamics* (improving/declining trajectories) rather than purely over levels.

Missingness handling. We exclude indicators with missingness above a threshold and drop rows that remain incomplete after feature construction (as specified in the configuration). The result is an engineered table $(\tilde{X}, y)$ to which we apply the same target-driven labeling and forest explanation procedures.

Algorithm 1 Original Method (Target-driven labeling)

Input: target vector $y \in \mathbb{R}^N$, quantile q
 Perform single-linkage hierarchical clustering on y
 Identify the largest cluster $C_{\max}$
 Initialize $z_i \leftarrow \mathbb{I}[i \notin C_{\max}]$
 Update $z_i \leftarrow z_i \vee \mathbb{I}[y_i \geq \text{Quantile}(y, q)]$
Output: labels $z \in \{0, 1\}^N$

3.3. Interpretable forest explanation

Given $(\tilde{X}, z)$, we learn compact decision-tree ensembles and select a forest that balances fidelity and simplicity.

Forest family. We search over many random seeds. For each seed s , we sample:

- number of trees $T_s \in \{1, \dots, 10\}$,
- per-tree question budget $Q_s \in \{2, \dots, 10\}$ (implemented as a max leaf-node constraint),

train T_s bootstrapped decision trees, prune pure subtrees, and aggregate predictions via an OR rule across trees. We record (i) training accuracy and (ii) total number of internal decision nodes (“questions”) across the forest. We then select a high-accuracy forest with minimal questions and export all trees (JSON + plots) for interpretability and auditing.

Interpretation note. The explanation model is intentionally constrained; it is not designed to maximize predictive performance on y , but to provide *human-readable* feature conditions that characterize the target-defined interesting set.

4. Algorithms

The pseudocode below matches the pipeline components used in experiments.

5. Experimental Setup
5.1. Datasets

Insurance (static). Medical Cost Personal dataset with target charges. We expect high-cost subgroups correlated with smoking status, age, and BMI.

California Housing (static). scikit-learn `fetch_california_housing` with target `MedHouseVal`. High-value subgroups are typically associated with coastal/urban regions and higher-income/household features.

Algorithm 2 SyFLoW-Inspired Method (Target-driven labeling)

Input: target y , seeds S , parameters $(\beta, \lambda, \rho, \delta, m_{\min}, \alpha_{\max})$
 Compute global histogram bins for y
 Initialize candidate set $\mathcal{C} \leftarrow \emptyset$
for $s = 1$ to S **do**
 Bootstrap sample $y^{(s)}$ from y
 Cluster $y^{(s)}$ with single-linkage to obtain clusters
 for each cluster c **do**
 Range $[c_{\min}, c_{\max}]$ from cluster extremal values
 Mask $m = \{i : c_{\min} \leq y_i \leq c_{\max}\}$
 Discard if $|m| < m_{\min}$ or $|m| > \alpha_{\max}N$
 quality(m) = $(|m|/N)^\beta \cdot \text{KL}(y_m \parallel y)$
 Add $(m, \text{quality}, y_m)$ to $\mathcal{C}$
 end for
end for
 Select $K = \max(1, \lfloor \rho|\mathcal{C}| \rfloor)$ candidates greedily:
 skip if Jaccard overlap with selected $> \delta$
 maximize quality(m) + $\lambda(|m|/N)^\beta \cdot \text{diversity}$
 Output z as the union of selected masks

Table 1. Target-driven labeling hyperparameters.

Component	Setting
Original method	high-value quantile $q = 0.94$
SyFLoW-inspired	number of bootstrap seeds $S = 1000$
	size exponent $\beta = 0.5$
	diversity weight $\lambda = 2.0$
	top candidate fraction $\rho = 0.01$
	max overlap $\delta = 1.0$
	min subgroup size $m_{\min} = 3$
	max subgroup fraction $\alpha_{\max} = 0.5$

World Bank Indicators (longitudinal). World Development Indicators panel (country $\times$ year) with target `life_expectancy_at_birth`. We build temporal features (lags, rolling statistics, trends) per country-year and apply the same subgroup-labeling and explanation pipeline.

5.2. Hyperparameters and configuration

We report all key parameters used in the experiments for reproducibility.

5.3. Evaluation outputs

We report: (i) KDE plots of the target distribution for interesting vs. non-interesting labels, (ii) forest accuracy versus total questions across seeds, (iii) exported tree visualizations, and (iv) for trend-target experiments, trend-slope distribution plots that visualize improving versus declining regimes. For the longitudinal dataset we additionally report an XG-

Algorithm 3 Longitudinal Feature Preparation

Input: panel data (e, t, X, y) , window w , missingness threshold τ , min history h
 Sort by entity e and time t
 Remove indicators with missing rate $> \tau$
for each entity e **do**
 for each time t with at least h observed target values **do**
 Compute lag-1, rolling mean/std, and trend for y over window w
 For each indicator, compute latest, lag-1, rolling mean/std, and trend over window w
 Emit a row $(\tilde{X}_{e,t}, y_{e,t})$
 end for
end for
 Drop rows with missing values (as configured)
Output: engineered table $(\tilde{X}, y)$

Algorithm 4 Forest Explanation for Subgroup Labels

Input: features $\tilde{X}$, labels z , seeds S
for $s = 1$ to S **do**
 Sample $T_s \in [1, 10]$, $Q_s \in [2, 10]$
 for $t = 1$ to T_s **do**
 Bootstrap rows from $(\tilde{X}, z)$
 Train a decision tree with max leaf nodes $\leq Q_s + 1$
 Prune pure subtrees
 end for
 Predict via OR aggregation across trees
 Record accuracy and total internal-node count (questions)
end for
 Select best forest: high accuracy, minimal questions
 Rebuild and export all trees for the best seed
Output: compact forest explanation and tree artifacts

Boost baseline with time-aware cross-validation.

6. Results

6.1. Static Data (2 Datasets, Original vs. SyFLoW-inspired)

We report (i) KDE plots comparing the target distribution of interesting ($z=1$) vs. non-interesting ($z=0$) instances, (ii) the accuracy–interpretability trade-off across candidate forests (accuracy vs. total decision questions), and (iii) the exported tree set for the selected compact forest.

6.2. Longitudinal Data (Level-target and Trend-target)

Reading guide. For longitudinal experiments we report results for (i) a **level target** ($y_{e,t}$) and (ii) a **trend target**

Table 2. Forest explanation hyperparameters.

Component	Setting
Forest search seeds	$S = 1000$
Accuracy quantile for filtering	0.99
Trees per forest	sampled uniformly from $\{1, \dots, 10\}$
Question budget	sampled uniformly from $\{2, \dots, 10\}$
Aggregation	OR across tree predictions

(local slope of y over a window). For each target mode, we apply the same target-driven labeling procedures (Original and SyFLoW-inspired) and learn an interpretable forest to explain the induced labels using the expanded temporal question vocabulary.

6.3. XGBoost Baseline

7. Discussion

What the pipeline discovers. Because subgroup identification is target-driven, the pipeline primarily detects instances with atypical target behavior (e.g., extreme values or distributional deviations). The explanation stage then translates these labels into concise feature rules. In the longitudinal setting, these rules are expressed in terms of recent dynamics (trends, lagged values, rolling variability), which offers a concrete interpretation of how the target behaves over time for the identified subgroup. In the longitudinal setting, we evaluate both a level-based target (life expectancy) and a trend-based target (local slope), enabling outcome-first subgrouping of both extreme outcomes and extreme trajectory changes.

Interpretability–fidelity trade-off. The forest search explicitly balances fidelity (accuracy on the subgroup label) against simplicity (total number of decision questions). This provides a tunable mechanism for producing explanations that are compact enough to be inspected, while still approximating the target-defined subgrouping.

Relationship to SyFLoW. Our SyFLoW-inspired method uses distributional exceptionality (KL divergence) and diversity to *propose* target-defined interesting sets. Unlike feature-space subgroup search methods, we do not directly optimize subgroup quality over feature predicates. Instead, we separate (i) target-based subgroup identification from (ii) feature-based explanation, which matches outcome-first analytic workflows.

Baseline interpretation. The XGBoost baseline measures the predictive signal of temporal features under time-aware cross-validation. It is not intended to replace interpretable subgroup rules; rather, it provides context on whether tem-

Table 3. Longitudinal configuration and XGBoost baseline settings (WBD).

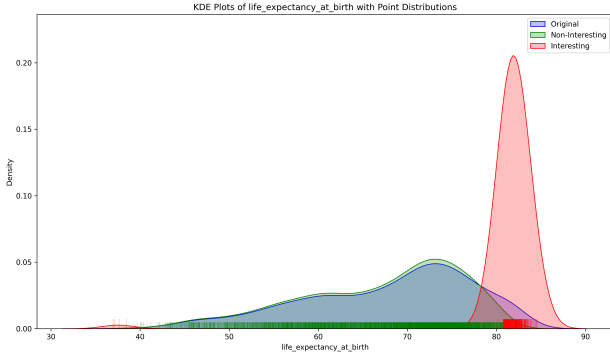
Component	Setting
Dataset	World Bank Development Indicators (country $\times$ year)
Panel mode	panel (stride = 1)
Window length	$w = 5$
Minimum history	$h = 3$
Max missingness per indicator/feature	0.5
Target	life.expectancy.at.birth
Target mode	trend (subgrouping on local slope over window w)
Target thresholds	{50, 70} (first crossing above/below features)
XGBoost CV	TimeSeriesSplit, $n_splits = 5$
XGBoost min samples	10
XGBoost params	$n_estimators = 1000$, $max_depth = 5$, $learning_rate = 0.05$, $subsample = 0.9$, $colsample_bytree = 0.9$, $objective=reg:squarederror$, $random_state = 42$

Table 4. XGBoost baseline summary on the WBD panel data (time-aware cross-validation).

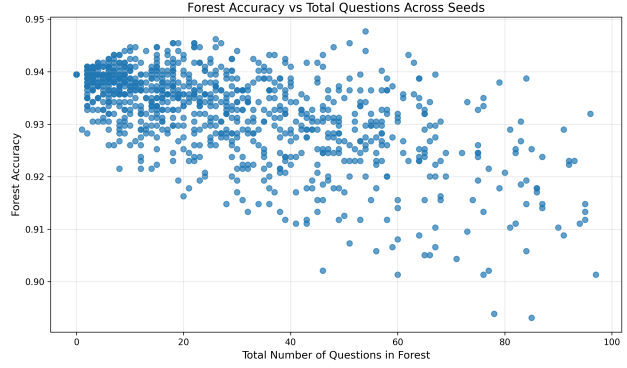
Metric	Mean	Std
RMSE	0.604	0.038
R^2	0.995	0.001

poral summaries contain strong predictive structure for the target.

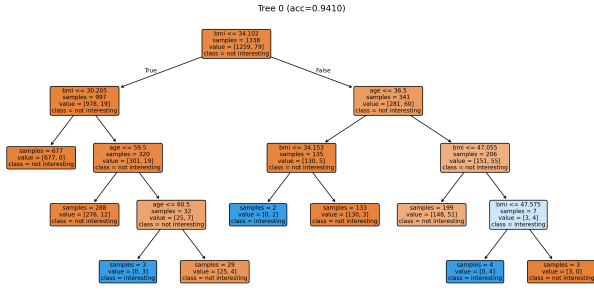
Limitations and next steps. First, target-driven labeling may select heterogeneous instances that share target behavior but not a single coherent feature description; this can force the explanation model into approximate rules. Second, our evaluation currently emphasizes fidelity and compactness; additional metrics such as subgroup stability across time, robustness to missingness, and external validity checks would strengthen conclusions. A natural extension is to integrate feature-space search (SyFLoW-style) directly on longitudinal features, enabling discovery of feature-defined subgroups with exceptional target distributions.



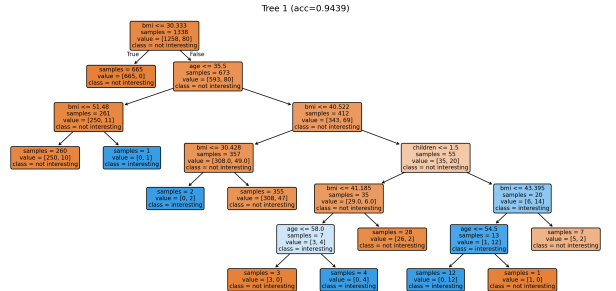
(a) Target separation (KDE): **Original labeling.**



(b) Forest search: accuracy vs. total questions.

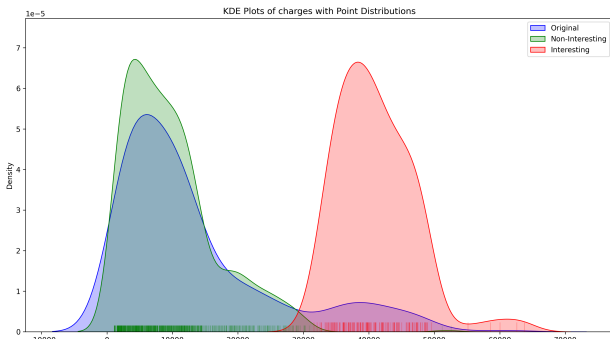


(c) Selected explanation forest: tree 1 (exported).

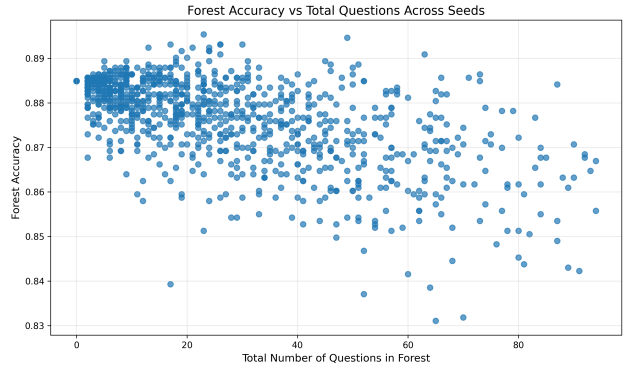


(d) Selected explanation forest: tree 2 (exported).

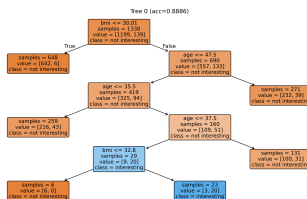
Figure 1. **Static dataset 1 (Insurance), Original labeling.** The KDE compares the target (charges) distribution for $z=1$ vs. $z=0$. The middle plot shows the accuracy–interpretability frontier explored during forest search, where “questions” are internal decision nodes. The bottom row shows exported trees from the selected compact forest.



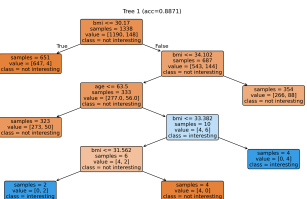
(a) Target separation (KDE): **SyFLoW-inspired labeling.**



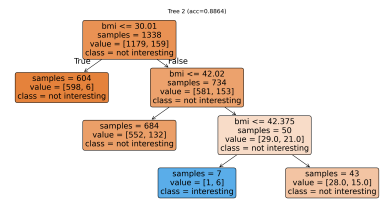
(b) Forest search: accuracy vs. total questions.



(c) Selected forest: tree A.



(d) Selected forest: tree B.



(e) Selected forest: tree C.

Figure 2. **Static dataset 1 (Insurance), SyFLoW-inspired labeling.** The KDE shows the target distribution split induced by the SyFLoW-inspired target-range selection. The forest search plot summarizes the accuracy vs. total questions trade-off; the selected solution is exported as a small set of trees.

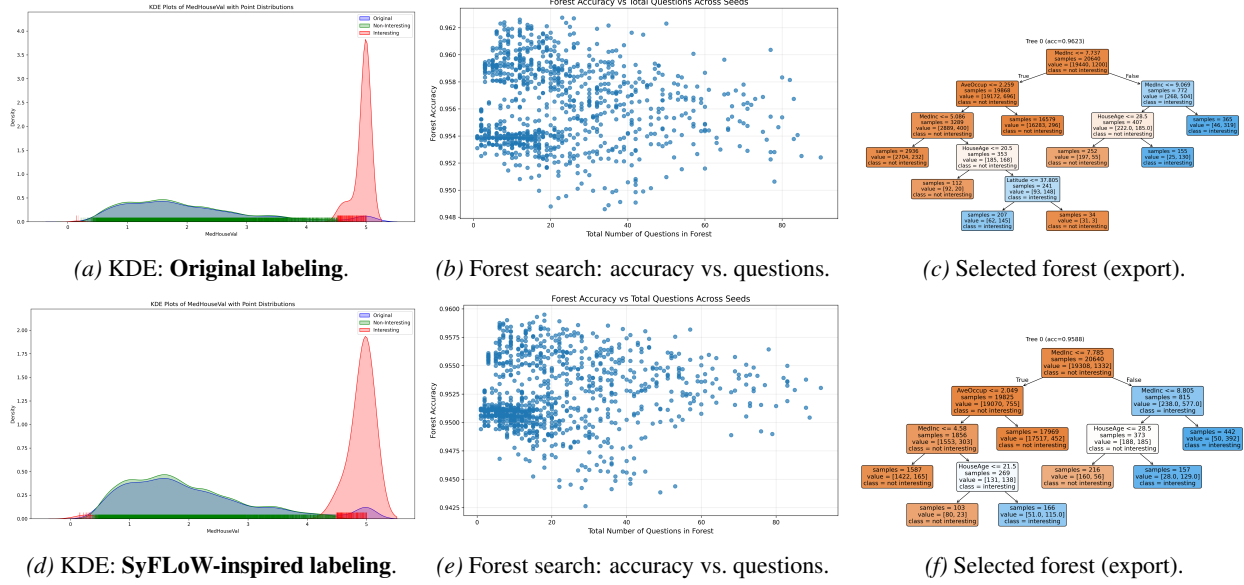


Figure 3. Static dataset 2 (California Housing): Original vs. SyFLoW-inspired labeling. Each row reports (left) KDE separation of the target (MedHouseVal) for $z=1$ vs. $z=0$, (middle) the forest search frontier (accuracy vs. total questions), and (right) the exported compact forest used for interpretation.

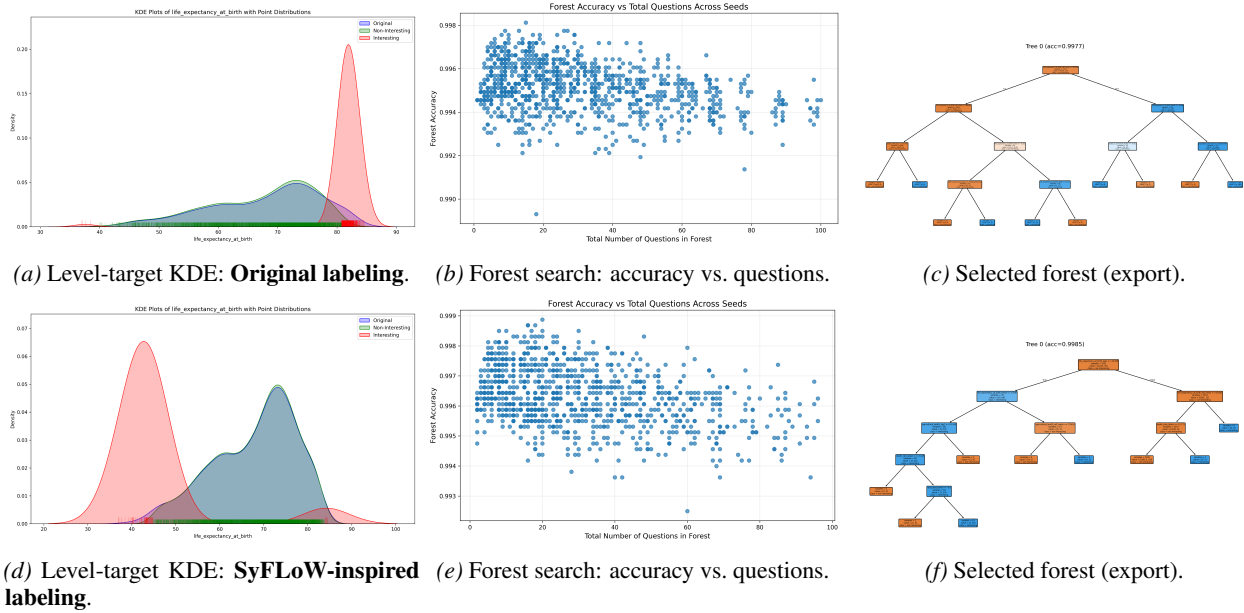


Figure 4. Longitudinal dataset 1 (WBD), level-target results. We subgroup country-year instances using the *level* of life expectancy as the target. For each labeling strategy (Original / SyFLoW-inspired), we show: KDE separation of the level target for $z=1$ vs. $z=0$; the accuracy-questions trade-off over candidate explanation forests; and the exported compact forest selected for interpretability. Explanation trees are trained on temporal descriptors (lags, deltas, rolling statistics, growth rates, regime shifts, and threshold-crossing times).

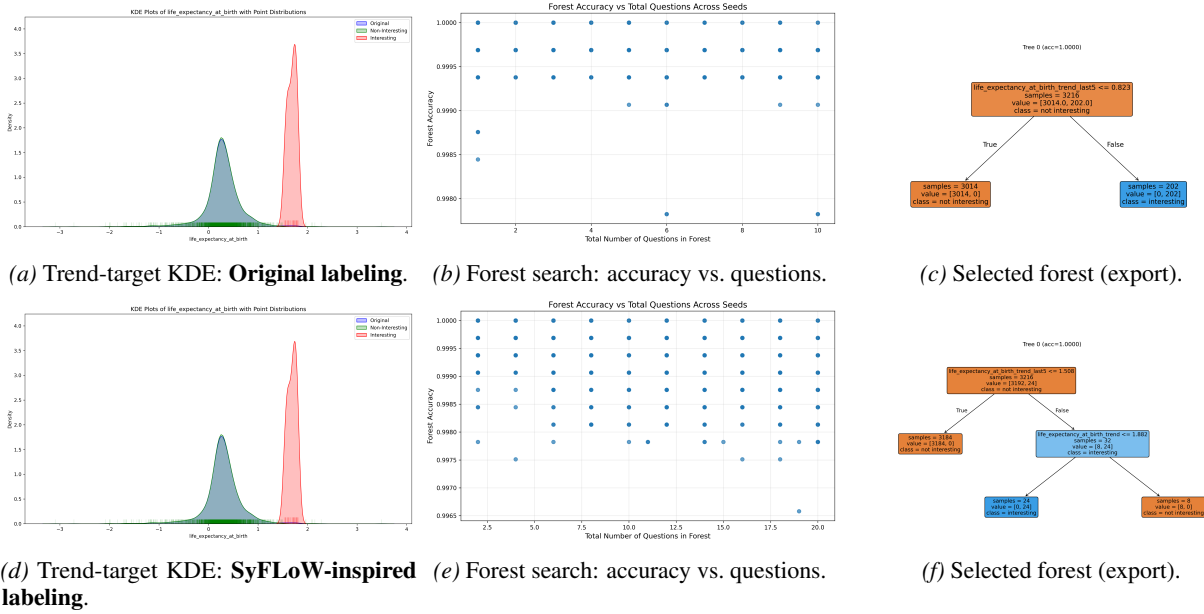


Figure 5. Longitudinal dataset 1 (WBD), trend-target results (new). Here the subgrouping target is the *local trend* (slope) of life expectancy computed over a rolling window. The KDE plots visualize separation between improving vs. declining regimes through the induced labels z . As in the level-target setting, we report the forest search frontier and export the selected compact explanation forest.